

Lei, G'Sell, Rinaldo, Tibshirani, Wasserman (2018)

Distribution-Free Predictive Inference for Regression

Paper Overview

1. Why this paper exists

From 2018 to 2022, this was *the* practitioner reference for conformal prediction (CP) in regression. Lei, G'Sell, Rinaldo, Tibshirani, and Wasserman Lei et al. (2018) adapted the abstract Vovk–Gammerman–Shafer framework Vovk, Gammerman, and Shafer (2005) to real-valued regression with a fitted point predictor $\hat{\mu} : \mathcal{X} \rightarrow \mathbb{R}$, producing three things the field needed.

Key Insight

What this paper unlocked. Vovk's original CP framework was algorithmic and very general, but it required full conformal prediction — refitting the underlying model once per candidate y . This is computationally infeasible for any modern regression model. Lei et al. showed that a much cheaper variant (*split conformal*) preserves the finite-sample marginal coverage guarantee exactly. This single observation turned CP from a theoretical curiosity into a deployable tool.

First, a formal proof that *split conformal regression* — where the data is partitioned into a training set used to fit $\hat{\mu}$ and a calibration set used to compute residuals — enjoys the same finite-sample marginal coverage guarantee as full conformal but at a fraction of the computational cost. Second, the *locally-weighted* (or σ -scaled) variant, which rescales calibration residuals by an estimate of the conditional spread $\hat{\sigma}(x)$ so that interval width

varies with x — a forerunner of the quantile-regression-based score in CQR Romano, Patterson, and Candès (2019). Third, the *LOCO* (leave-one-covariate-out) notion of model-free variable importance, which sits independently of the entire selective-inference / debiased-lasso machinery.

Plain English

Three concrete contributions, in order of practical impact:

1. Split conformal regression: marginal coverage at the cost of one model fit (instead of one fit per test point).
2. σ -scaled (locally adaptive) variant: intervals that widen in noisy regions and narrow in quiet ones.
3. LOCO: a way to ask “does feature j matter?” without committing to any model class.

The accompanying R package `conformalInference` bundled all three and was for years the dominant CP-in-R implementation.

2. Background and terminology

A *predictor* $\hat{\mu} : \mathcal{X} \rightarrow \mathbb{R}$ is any regression function fitted on training data — the paper treats $\hat{\mu}$ as a *black box* and inspects only its outputs. The data are assumed *exchangeable*: any reshuffling produces the same joint distribution. This is strictly weaker than i.i.d., though i.i.d. data are exchangeable.

What This Definition Means

Exchangeability is the only thing the proof needs. It is strictly weaker than i.i.d. (which requires both identical distribution AND independence). Practical consequence: any random reshuffling of your data yields an exchangeable sequence, so as long as you train/calibrate/test on randomly-split data you can apply CP. Time-ordered splits break exchangeability — a separate body of work (Stocker et al., Zaffran et al.) handles that case.

A *calibration set* is a held-out subset $\{(X_i, Y_i)\}_{i=n_1+1}^{n_1+n_2}$ disjoint from the training set used to fit $\hat{\mu}$. The split-conformal variant produces its cutoff from this set alone, so it never has to refit $\hat{\mu}$ at test time. The *miscoverage level* $\alpha \in (0, 1)$ is the maximum acceptable failure probability; setting $\alpha = 0.1$ asks for 90% prediction intervals.

3. The constructions

Split conformal regression

Given a training set, a calibration set $\{(X_i, Y_i)\}_{i=1}^n$ disjoint from it, a fitted predictor $\hat{\mu}$, and a target miscoverage α :

Step 1: residuals on calibration. Compute $R_i = |Y_i - \hat{\mu}(X_i)|$ for $i = 1, \dots, n$.

Step 2: adjusted-level quantile.

$$\underbrace{\hat{q}}_{\text{cutoff that calibrates the interval}} = \text{Quantile}\left(\underbrace{\{R_i\}_{i=1}^n}_{\text{absolute residuals on held-out data}} ; \underbrace{\frac{\lceil (n+1)(1-\alpha) \rceil}{n}}_{\text{slightly above } 1-\alpha; \text{ the } +1 \text{ corrects for the unseen test point}} \right).$$

Step 3: prediction interval. For a fresh input X_{n+1} ,

$$\underbrace{\hat{C}(X_{n+1})}_{\text{prediction interval handed to the user}} = \left[\underbrace{\hat{\mu}(X_{n+1}) - \hat{q}}_{\text{lower endpoint}}, \underbrace{\hat{\mu}(X_{n+1}) + \hat{q}}_{\text{upper endpoint}} \right].$$

How This Works

The recipe in three sentences. Hold out a calibration set. Take absolute residuals on that set. Quantile them at the slightly-adjusted level $\lceil (n+1)(1-\alpha) \rceil / n$ and use that as a $\pm$ band around any prediction. The model $\hat{\mu}$ is never inspected, never refit, never modified — it is wrapped from the outside.

What changed from full conformal. The full-conformal construction (originally in Vovk, Gammerman, and Shafer (2005)) requires refitting $\hat{\mu}$ once for every candidate value $y \in \mathbb{R}$ to test whether y belongs to the prediction set. This is computationally infeasible for general regression. The split variant refits $\hat{\mu}$ *zero* extra times: the price is that calibration data must be set aside and cannot be re-used for training.

Locally-weighted (σ -scaled) conformal regression

A constant cutoff $\hat{q}$ produces intervals of constant width $2\hat{q}$. This is wasteful when the conditional variance $\text{Var}(Y \mid X = x)$ varies substantially across $\mathcal{X}$ — a typical setting being heteroscedastic financial returns or seasonal energy demand. The paper proposes rescaling residuals by an estimate $\hat{\sigma}(x)$ of conditional spread (e.g., a separately fitted scale

function):

$$\underbrace{R_i}_{\text{score}} = \frac{\underbrace{|Y_i - \hat{\mu}(X_i)|}_{\text{residual}}}{\underbrace{\hat{\sigma}(X_i)}_{\text{local scale at } X_i}}.$$

The resulting prediction interval becomes

$$\hat{C}^\sigma(X_{n+1}) = \left[\hat{\mu}(X_{n+1}) - \underbrace{\hat{q} \hat{\sigma}(X_{n+1})}_{\text{locally rescaled half-width}}, \hat{\mu}(X_{n+1}) + \hat{q} \hat{\sigma}(X_{n+1}) \right],$$

so the width is now $2\hat{q} \hat{\sigma}(X_{n+1})$ — narrow where $\hat{\sigma}$ is small, wide where it is large. CQR Romano, Patterson, and Candès (2019) later generalises this from a separately fitted scale function to a fully quantile-regression-based predictor.

Key Insight

Why σ -scaling matters. Constant-width intervals are statistically valid but practically wasteful: they're too wide for easy inputs and too narrow for hard ones. The σ -scaled variant gets the same marginal coverage guarantee *plus* a degree of conditional adaptivity — intervals that look right at each input. CQR generalises the trick: instead of estimating the conditional scale, estimate the conditional quantiles directly.

Jackknife predictive intervals

The paper also analyses the leave-one-out (jackknife) construction: fit $\hat{\mu}_{-i}$ on the data with (X_i, Y_i) held out, compute the LOO residual $R_i^{\text{LOO}} = |Y_i - \hat{\mu}_{-i}(X_i)|$, and form

$$\hat{C}^{\text{jack}}(X_{n+1}) = \left[\hat{\mu}(X_{n+1}) - \hat{q}^{\text{LOO}}, \hat{\mu}(X_{n+1}) + \hat{q}^{\text{LOO}} \right],$$

where $\hat{q}^{\text{LOO}}$ is the quantile of the LOO residuals. The catch: jackknife coverage is established only *asymptotically*, under stability/MSE assumptions on the base estimator $\hat{\mu}$. This is strictly weaker than split conformal's finite-sample distribution-free guarantee. The Jackknife+ construction of Barber, Candès, Ramdas, and Tibshirani Barber et al. (2021) later closes this gap, recovering a finite-sample distribution-free $1 - 2\alpha$ guarantee.

What This Assumption Means

Why the jackknife is weaker than split conformal here. Split conformal works at finite n , distribution-free, with only exchangeability needed. The plain jackknife adds an *asymptotic* guarantee and requires the base estimator to satisfy a stability condition (small changes in training data $\rightarrow$ small changes in predictions). In high-dimensional or

modern-ML settings, these stability conditions are usually not verifiable. The takeaway: prefer split conformal unless you have a specific reason for the jackknife.

4. The coverage results

THEOREM (Marginal coverage, split conformal). *If $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable, then the split-conformal interval $\hat{C}(X_{n+1})$ satisfies*

$$\underbrace{\mathbb{P}(Y_{n+1} \in \hat{C}(X_{n+1}))}_{\substack{\text{probability the true label} \\ \text{lies in the interval}}} \geq \underbrace{1 - \alpha}_{\text{nominal level}}.$$

The upper bound $1 - \alpha + 1/(n + 1)$ holds under the additional assumption that the calibration scores have no ties.

What This Result Says

The shape of the guarantee in this paper:

- Finite-sample (works for any n , not just large).
- Distribution-free (no assumption on the law of (X, Y)).
- Model-agnostic (no assumption on $\hat{\mu}$).
- Marginal (averaged over the joint draw of test point and calibration set).
- Only *lower*-bounded: actual coverage is between $1 - \alpha$ and $1 - \alpha + 1/(n + 1)$.

Proof intuition. Under exchangeability of $(X_i, Y_i)_{i=1}^{n+1}$, the rank of the unobserved test residual $|Y_{n+1} - \hat{\mu}(X_{n+1})|$ among the $n + 1$ residuals (the n calibration residuals plus the test residual) is uniformly distributed on $\{1, \dots, n + 1\}$. Setting the cutoff at the $\lceil (n + 1)(1 - \alpha) \rceil$ -th smallest calibration residual exactly bounds the probability that the test rank exceeds the cutoff. The σ -scaled variant has the same proof: rescaling preserves exchangeability of the scores.

Oracle bounds. The paper goes further than marginal coverage. Under stability assumptions on $\hat{\mu}$, Theorems 3.2 and 3.3 of Lei et al. (2018) bound the symmetric difference between the conformal interval and the *oracle band* (the narrowest band achieving the same coverage at the population level). Under consistency of $\hat{\mu}$ (Theorems 3.4 and 3.5), the conformal interval approaches the *super-oracle* (the narrowest band achieving conditional coverage). These results explain why the locally-weighted variant works in practice: when $\hat{\sigma}$ is consistent for the conditional spread, the σ -scaled interval converges to the oracle

locally adaptive band.

Plain English

What the oracle bounds give you that the basic coverage theorem doesn't. Marginal coverage says the interval is valid *on average*. The oracle bounds say something stronger: when the base model is well-fit, the conformal interval is also *narrow* — close to the narrowest possible interval that achieves the same coverage. So a well-calibrated σ -estimator yields intervals that are not just valid but also close-to-optimal in width.

Rank-One-Out (ROO).. A subsidiary contribution: an extension that produces valid *in-sample* prediction intervals (intervals around each training point's own label) at essentially the cost of split conformal. Useful for diagnostics and residual analysis.

5. LOCO: model-free variable importance

The paper additionally introduces *LOCO* (leave-one-covariate-out) as a distribution-free notion of variable importance. For covariate j , refit $\hat{\mu}$ on the dataset with X_j removed; the increase in prediction error is the LOCO importance of j . The construction is intentionally model-free: it requires neither linearity, sparsity, nor exchangeability beyond what conformal coverage already needs. LOCO inference exploits the same rank-based ideas as the prediction-set construction, producing valid finite-sample confidence intervals for the importance score under no further assumptions on $\hat{\mu}$.

Key Insight

LOCO is the spiritual ancestor of permutation importance and SHAP. It asks “how much worse does my model do if I throw away feature j ?” — and gives a distribution-free confidence interval for that answer. Unlike SHAP (which decomposes a fitted model's predictions) or selective inference (which requires linearity assumptions), LOCO works for any black-box $\hat{\mu}$ and any data distribution.

6. Limits, extensions, and connections

The construction has the same fundamental limit as all distribution-free CP: marginal coverage is not conditional coverage. Lei and Wasserman (2014) established the formal X -conditional impossibility result in 2014: distribution-free conditional coverage at non-trivial level is achievable only by sets of infinite Lebesgue measure. The paper's locally-weighted variant is an *approximate* approach to this problem — when $\hat{\sigma}$ is

well-estimated, the marginal-coverage interval is also approximately conditionally valid (the formal sense is the super-oracle bound).

Why This Matters

The honest limit you should know about. CP cannot give you *per-point* coverage guarantees in the distribution-free setting — the Lei-Wasserman result rules it out. The σ -scaled variant is the practitioner’s best honest answer: it tracks the conditional spread approximately, so the marginal-coverage interval is also *approximately* conditionally valid when your scale estimator is good. If you need genuine per-point guarantees, you must give up distribution-freeness and assume something specific about $P(Y | X)$.

The jackknife analysis identifies a weakness that the field would spend the next three years closing. The Jackknife+ paper Barber et al. (2021) proves a finite-sample distribution-free $1 - 2\alpha$ guarantee using a comparison-matrix (tournament) argument, and extends to K -fold CV+. The same authors later generalise CP to non-exchangeable data via fixed weights, building on the weighted-CP construction of Tibshirani et al. Tibshirani et al. (2019). The lineage from this paper through these extensions is recognisable.

The other reader’s guides in this collection extend the foundation in specific directions. Angelopoulos and Bates Angelopoulos and Bates (2022) is the modern ML-practitioner companion to this paper. Romano, Patterson, and Candès Romano, Patterson, and Candès (2019) replace the σ -scaled residual with a full quantile-regression score, producing CQR. The Tibshirani 2023 CMU lecture notes give the statistician’s-eye view of the same construction, with explicit Beta-distributed calibration-conditional coverage analysis and a careful presentation of the Lei–Wasserman impossibility.

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